

Mathew Terhune

[MATHEWTERHUNE.ME](https://mathefterhune.me) | [LINKEDIN](#) | [GITHUB](#) | MATHEWTERHUNE@GMAIL.COM

EDUCATION

University of Victoria
BSc in Computer Science

Sep. 2021 – April 2025
Victoria, BC, Canada

RESEARCH EXPERIENCE

Cortex Labs: DECISION-MAP: [WEBSITE](#)
Undergraduate research assistant & developer

Sep. 2023 – April 2025
Victoria, BC, Canada

- Designed and helped maintain a visualization tool relating to brain injury, mental health, and addiction.
- Project has been presented globally at events including Canadian MP, Canadian Traumatic Brain Injury Research Consortium, and the Global Neuropsychology Congress in Portugal.
- Designed landing website for the project. (no longer in use)

PROFESSIONAL EXPERIENCE

Columbia Basin Trust
CO-OP, Information Systems

May 2024 – Sep. 2024
Castlegar, BC, Canada

- Responded to and resolved support tickets, ensuring timely and effective troubleshooting of technical issues.
- Researched, tested, documented, presented, and prototyped the integration of Microsoft Power BI to management.
- Assisted in regular servicing and secure data wiping on physical servers.
- Managed and maintained multiple virtual machines, creating a virtual network for communication.

HIGHLIGHTED PROJECTS

Graph Search Algorithm Visualization Tool: [GITHUB](#)

Ongoing

- Developed interactive graph visualization tool using React, allowing real-time exploration of algorithms such as BFS & DFS
- Implemented dynamic grid resizing, node, selection, and visualization controls optimized with React state management and event handling.

Search Algorithm Comparison in Decentralized P2P Networks: [WEBSITE](#) | [GITHUB](#)

Spring 2025

- Designed and implemented a custom Gnutella-like network simulation using C++ NS3 to evaluate search algorithms.
- Compared flooding, normalized flooding, and k-random walks across several different graph types.
- Created custom packet headers, topology, builders, and forwarding logic to simulate real life node behaviour and track metrics.

Parallel Pathfinding with A-Star: [GITHUB](#)

Fall 2024

- Designed a GPU-optimized priority queue with CUDA to be used within a parallel implementation of A-star pathfinding algorithm.
- Profile GPU kernels using NVIDIA's NCU-CLI, revealing a 90%+ DRAM throughput with memory bound workloads
- Constructed and manipulated unit grid graphs with adjustable obstacle configuration for benchmarking traversal performance.
- Accompanied with a visualization tool created in python.

NHL Game Outcome Predictor: [WEBSITE](#)

Spring 2024

- Built real-time game prediction model using 1.75M+ NHL shot events from MoneyPuck.com.
- Leveraged machine learning tactics to predict the outcome of NHL games based on historical shot data.
- Engineered shot quality features for logistic regression, then integrated into a LSTM to predict outcomes dynamically.

EduPath – Educational Pathway Planner: *(Private repository, available on request)*

Ongoing

- Full-stack application that helps students explore programs, compare institutions, and assess their eligibility to enroll.
- Designed an interactive React-based website allowing students to browse institutions and search for academic programs
- Built and deployed a PostgreSQL database using Supabase to manage institutions, programs, and pre-requisites, with secure API endpoints enabling dynamic retrieval from the front-end.

TECHNICAL SKILLS

Languages: C++, C, Python, JavaScript, CUDA, Flutter, Haskell, Rust.

Back-End: MySQL, PostgreSQL, Node.js, Express.js.

Front-End: HTML, CSS, JavaScript, React.js, Vue.js, tailwind.css.

Operating Systems: Unix/Linux, Windows, Mac OSX.

Software: Git, Microsoft Teams, Lists, Planner, Power BI, Word, Excel, Visual Studio Code, IntelliJ, Xcode, Wireshark.